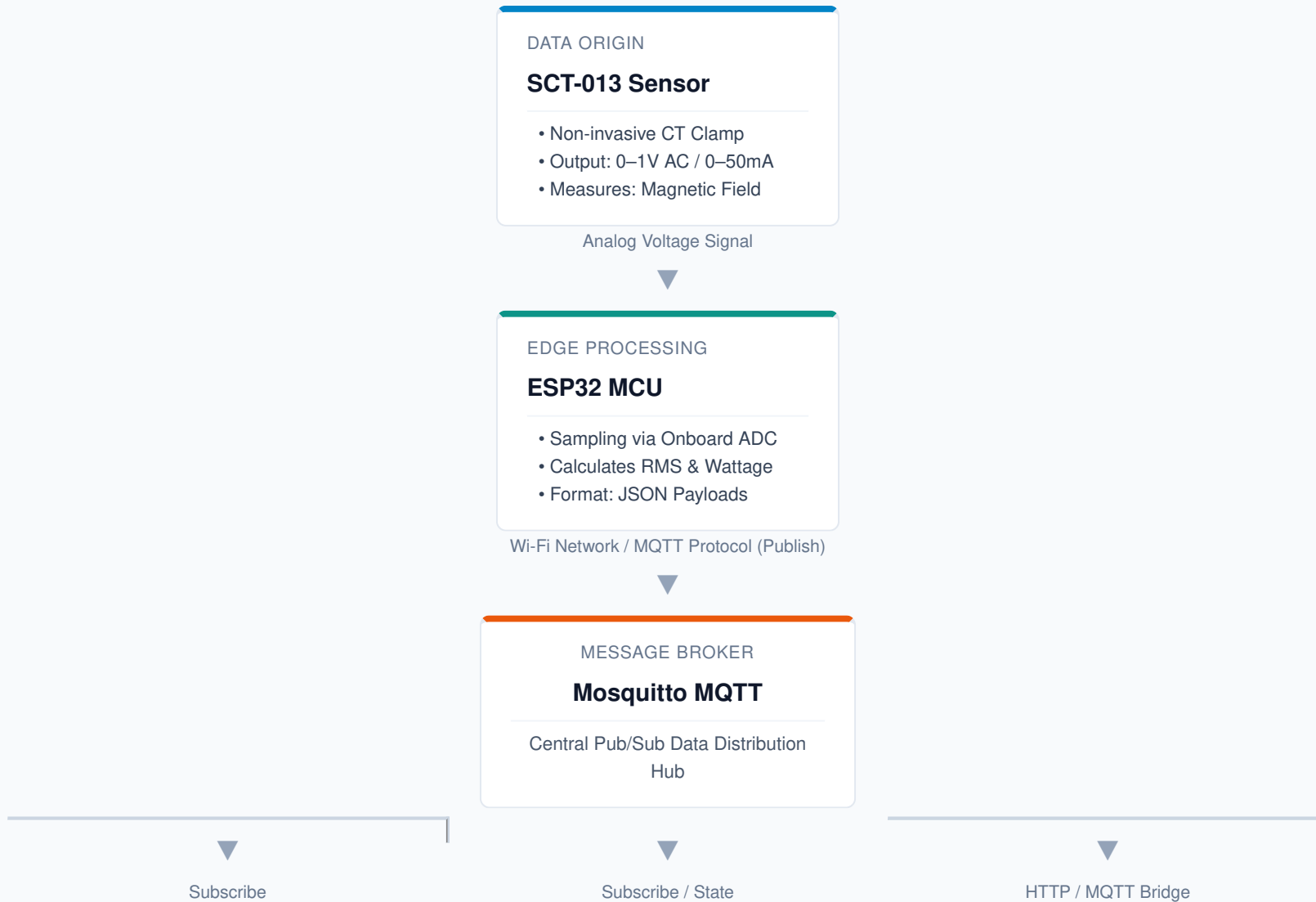


IoT Current Monitoring Architecture

Clean Data Flow Diagram: From Physical Measurement to Analytics & Automation Ecosystem



TIME-SERIES ANALYTICS

Grafana

- Data Source: InfluxDB / TSDB
- Historical trend graphs
- Long-term metric analysis

HOME AUTOMATION

Home Assistant

- Real-time dashboard UI
- Automations & Thresholds
- Energy Management tab

CLOUD IOT ANALYTICS

ThingSpeak

- Remote cloud storage
- MATLAB visualization
- Public / Private feeds

Data Pipeline Technical Overview

Pipeline Stage	Mechanism & Transformation	Payload / Protocol Format
Sensor → ESP32	Continuous AC current sampling across a burden resistor to transform current induced in secondary coil into proportional safe voltage.	Analog Voltage (Continuous Sine Wave form) sampled natively via ESP32 ADC pin.
ESP32 Processing	Calculates root-mean-square (RMS) current over 1000+ cycles, multiplies by line voltage to compute apparent or real power.	Internal float conversion variables transformed into string notation packets.
ESP32 → Broker	WiFi connection establishes TCP connection with Mosquitto, publishing readings at periodic intervals (e.g., every 5-10s).	MQTT JSON payload on topic tele/ems/current: { "Amps": 4.15, "Watts": 954.5, "Uptime": 3600 }
Broker → Consumers	Mosquitto routes data asynchronously to targets matching subscribed topic filters. Telegraf/Node-RED stores to Database for Grafana.	Lightweight standard MQTT Pub/Sub messaging overhead over standard ports 1883 or 8883 (TLS).